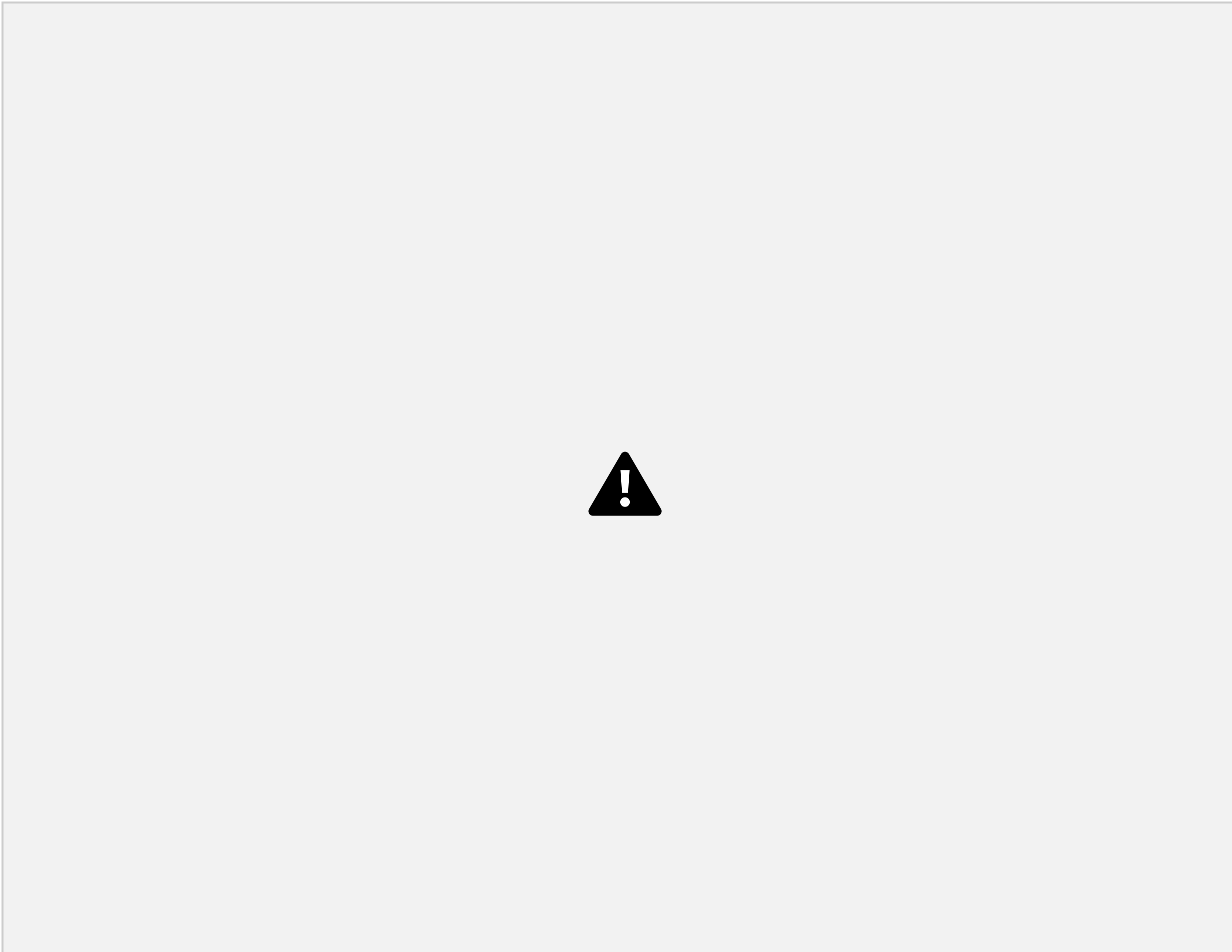
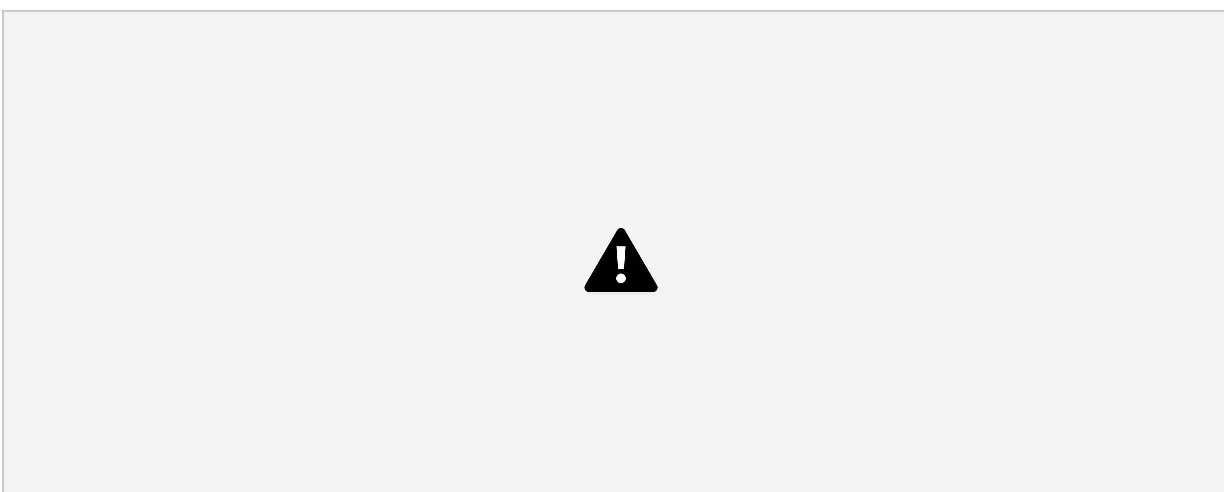
# Numerical Integration

## Trapezoidal Method





## Pseudocode

### 1. Start of the Program

- Begin by defining two function prototypes:

- `F(float x)`: A function that computes a value based on `x`. (Note: The exact definition is not shown in the code.)
- `trapzd(float a, float b, int n)`: A function that will calculate the numerical integral of function `F` over the interval [a, b] using the trapezoidal rule.

**2. Main Function:**

- Declare the following variables:
  - `int n`: The number of intervals to divide the integration range into.
  - `float a, b`: These will represent the lower and upper limits of the integration.
  - `float s`: This will store the result of the integration.
- Ask the user to input the values of `a`, `b`, and `n` (the limits of integration and the number of intervals).
- Call the function `trapzd(a, b, n)` with the provided input values, and store the result in `s`.
- Display the value of `s`, which represents the numerical integral of the function `F(x)` over the interval [a, b].

**3. Function: `trapzd(float a, float b, int n)`**

- **Input:**
  - `a`: The lower limit of integration.
  - `b`: The upper limit of integration.
  - `n`: The number of intervals to divide the range [a, b].
- **Output:**
  - Returns the estimated value of the integral using the trapezoidal rule.
- **Process:**
  - Declare the following variables:
    - `float x`: A temporary variable to store the current point on the x-axis.
    - `float tnm`: This variable is unused in the current code.
    - `float s`: This will store the current approximation of the integral.
    - `float sum`: This will accumulate the sum of function values at internal points between `a` and `b`.
    - `float del`: This represents the width of each sub-interval and is calculated as `(b - a) / n`.
  - Calculate the initial value of `s` as `0.5 * del * (F(a) + F(b))`, which is based on the endpoints of the interval.
  - Set `sum` to 0, to accumulate intermediate sums.
  - **Check if `n == 1`:**
    - If `n` is 1, it means the interval is divided into one trapezoid, so return the value of `s` as the result.
  - **If `n > 1`:**
    - Loop through all the internal points from `j = 1` to `j = n - 1` (i.e., excluding the endpoints `a` and `b`).
    - For each `j`, calculate the corresponding `x` value using the formula: `x = a + j * del`.
    - Accumulate the function value `F(x)` into `sum`.

- After the loop, update the integral value `s` by adding `del * sum` (the contribution from the internal points).
- Return the final value of `s`.